

Class 18: Pertussis Vaccination Mini Project

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Investigating pertussis cases by year

Pertussis (a.k.a. Whooping Cough) is a deadly lung infection caused by the bacteria B. Pertussis.

The CDC tracks Pertussis cases around the U.S. <http://tinyurl.com/pertussiscdc>

Q1. With the help of the R “addin” package `datapasta` assign the CDC pertussis case number data to a data frame called `cdc` and use `ggplot` to make a plot of cases numbers over time.

We can “scrape” this data using the R **`datapasta`** package.

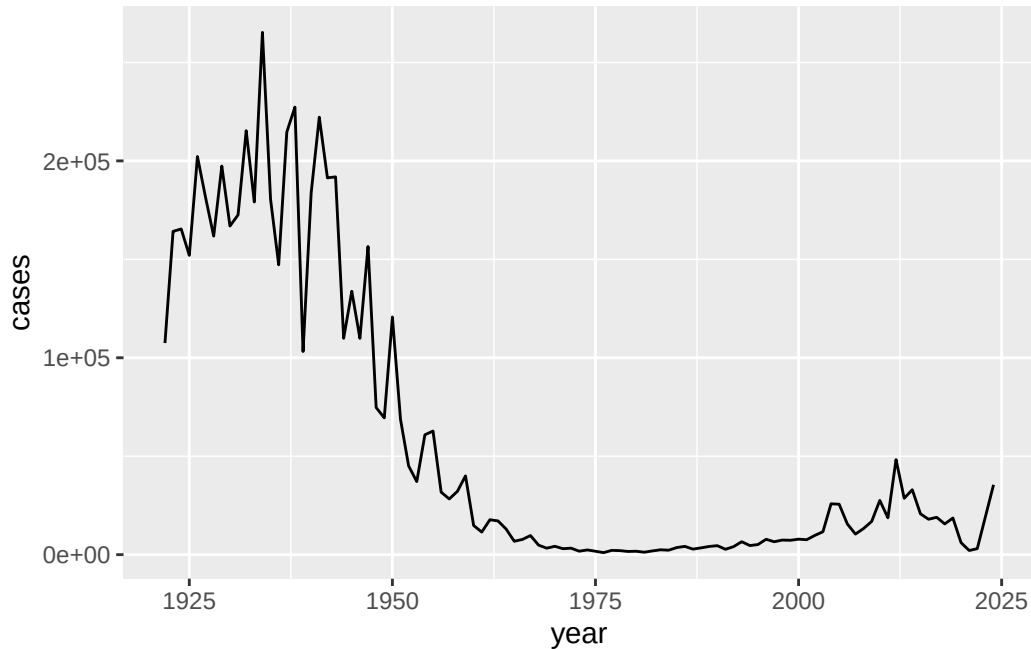
```
head(cdc)
```

```
  year  cases
1 1922 107473
2 1923 164191
3 1924 165418
4 1925 152003
5 1926 202210
6 1927 181411
```

A tale of two vaccines (wP and aP)

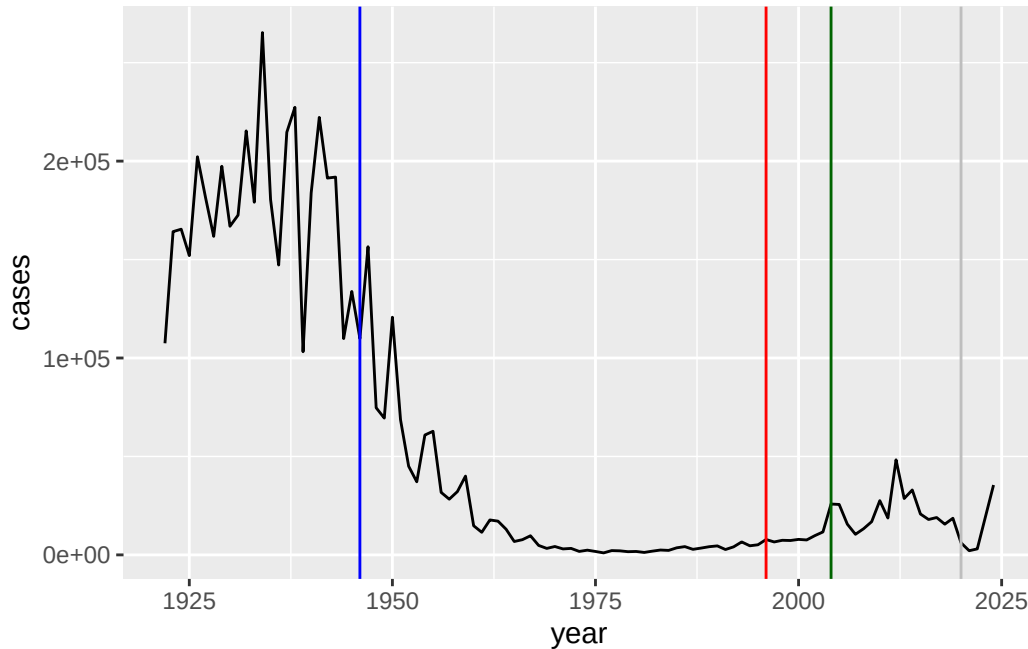
```
library(ggplot2)
```

```
ggplot(cdc) +  
  aes(x = year, y = cases) +  
  geom_line()
```



Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc) +  
  aes(x = year, y = cases) +  
  geom_line() +  
  geom_vline(xintercept = 1946, col = "blue") +  
  geom_vline(xintercept = 1996, col = "red") +  
  geom_vline(xintercept = 2020, col = "gray") +  
  geom_vline(xintercept = 2004, col = "darkgreen")
```



There were high case numbers before the first wP (whole cell) vaccine rollout in 1946. Then there was a rapid decline in case numbers until 2004 (4 years after aP vaccine rollout) when we have our first large-scale outbreak of pertussis again. There is also a notable COVID related dip and recent rapid rise.

Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

CDC data shows a resurgence of pertussis, with 48,277 cases reported in 2012 which is the highest levels since 1955. Possible causes include improved PCR testing, vaccine hesitancy, bacterial evolution, and waning immunity from the newer aP vaccine. The aP vaccine only has the antigens of the bacteria instead of the “whole-cell” inactivated bacteria, so we can do more analysis to see what is different about the immune response to infection in patients that had the older wP vaccine compared to the newer aP vaccine.

3. Exploring Computational Models of Immunity Pertussis Boost (CMI-PB) Data

The CMI-PB project aims to address this key question: what is different between aP and wP individuals.

We can get all the data from this ongoing project via JSON API calls. For this we will use the `jsonlite` package. We can install with `install.packages("jsonlite")`

```
library(jsonlite)
```

Warning: package 'jsonlite' was built under R version 4.4.3

```
subject <- read_json("https://www.cmi-pb.org/api/v5_1/subject", simplifyVector = TRUE)
head(subject)
```

	subject_id	infancy_vac	biological_sex	ethnicity	race
1	1	wP	Female	Not Hispanic or Latino	White
2	2	wP	Female	Not Hispanic or Latino	White
3	3	wP	Female	Unknown	White
4	4	wP	Male	Not Hispanic or Latino	Asian
5	5	wP	Male	Not Hispanic or Latino	Asian
6	6	wP	Female	Not Hispanic or Latino	White

	year_of_birth	date_of_boost	dataset
1	1986-01-01	2016-09-12	2020_dataset
2	1968-01-01	2019-01-28	2020_dataset
3	1983-01-01	2016-10-10	2020_dataset
4	1988-01-01	2016-08-29	2020_dataset
5	1991-01-01	2016-08-29	2020_dataset
6	1988-01-01	2016-10-10	2020_dataset

How many individuals “subjects” are in this dataset?

```
nrow(subject)
```

```
[1] 172
```

172 individuals

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```
aP wP
87 85
```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```
Female    Male  
    112     60
```

Q6. What is the breakdown of race and biological sex?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

```
#produces a cross-table
```

Not representative of the US population (more representative of UCSD population), so the conclusions are not going to be representative.

Side-Note: Working with dates

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

```
date, intersect, setdiff, union
```

What is today's date?

```
today()
```

```
[1] "2025-03-08"
```

```
today() - ymd("2000-01-01")
```

Time difference of 9198 days

What is this in years?

```
time_length( today() - ymd("2000-01-01"), "years")
```

```
[1] 25.18275
```

we are using the `ymd()` function to tell lubridate the format of our particular date and then the `time_length()` function to convert days to years.

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
# Use today's date to calculate age in days
subject$age <- today() - ymd(subject$year_of_birth)
```

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
ap <- subject %>% filter(infancy_vac == "aP")
round( summary( time_length( ap$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22	26	27	27	28	34

```
# wP
wp <- subject %>% filter(infancy_vac == "wP")
round( summary( time_length( wp$age, "years" ) ) )
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
22	32	34	36	39	57

Q8. Determine the age of all individuals at time of boost?

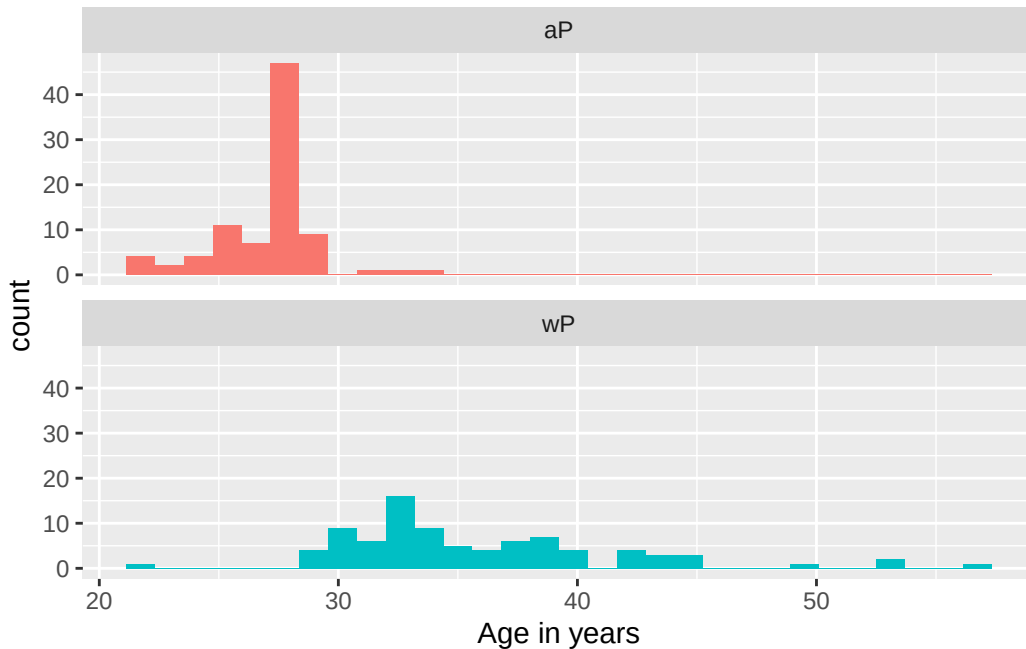
```
int <- ymd(subject$date_of_boost) - ymd(subject$year_of_birth)
age_at_boost <- time_length(int, "year")
head(age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"),
      fill=as.factor(infancy_vac)) +
  geom_histogram(show.legend=FALSE) +
  facet_wrap(vars(infancy_vac), nrow=2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value with `binwidth`.



The two groups look different with the age of the individuals that took the aP vaccine being overall younger. We can also calculate a p-value to determine significance.

```
# Or use wilcox.test()
x <- t.test(time_length( wp$age, "years" ),
            time_length( ap$age, "years" ))

x$p.value
```

```
[1] 2.372101e-23
```

Joining multiple tables

Obtain more data from CMI-PB:

```
specimen <- read_json("http://cmi-pb.org/api/v5_1/specimen", simplifyVector = T)

ab_data <- read_json("http://cmi-pb.org/api/v5_1/plasma_ab_titer", simplifyVector= T)

head(specimen)
```

	specimen_id	subject_id	actual_day_relative_to_boost	
1	1	1	-3	
2	2	1	1	
3	3	1	3	
4	4	1	7	
5	5	1	11	
6	6	1	32	

	planned_day_relative_to_boost	specimen_type	visit
1	0	Blood	1
2	1	Blood	2
3	3	Blood	3
4	7	Blood	4
5	14	Blood	5
6	30	Blood	6

```
head(ab_data)
```

	specimen_id	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	1	IgE	FALSE	Total	1110.21154	2.493425
2	1	IgE	FALSE	Total	2708.91616	2.493425
3	1	IgG	TRUE	PT	68.56614	3.736992
4	1	IgG	TRUE	PRN	332.12718	2.602350
5	1	IgG	TRUE	FHA	1887.12263	34.050956
6	1	IgE	TRUE	ACT	0.10000	1.000000

	unit	lower_limit_of_detection
1	UG/ML	2.096133
2	IU/ML	29.170000
3	IU/ML	0.530000
4	IU/ML	6.205949
5	IU/ML	4.679535
6	IU/ML	2.816431

I now have three tables of data from CMI-PB: `subject`, `specimen`, and `ab_data`. I need to “join” these tables so I will have all the info I need to work with.

For this we will use the `inner_join()` function from **dplyr** package.

Q9. Complete the code to join `specimen` and `subject` tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
library(dplyr)
```

```
meta <- inner_join(specimen, subject)
```

Joining with `by = join\_by(subject\_id)`

```
head(meta)
```

	specimen_id	subject_id	actual_day_relative_to_boost
1	1	1	-3
2	2	1	1
3	3	1	3
4	4	1	7
5	5	1	11
6	6	1	32

	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex
1	0	Blood	1	wP	Female
2	1	Blood	2	wP	Female
3	3	Blood	3	wP	Female
4	7	Blood	4	wP	Female
5	14	Blood	5	wP	Female
6	30	Blood	6	wP	Female

	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	age
1	14311 days
2	14311 days
3	14311 days
4	14311 days
5	14311 days
6	14311 days

```
dim(subject)
```

```
[1] 172  9
```

```
dim(specimen)
```

```
[1] 1503    6
```

```
dim(meta)
```

```
[1] 1503   14
```

We can join our `ab_data` table to `meta` so we have all the info we need about antibody levels.

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(meta, ab_data)
```

Joining with ``by = join_by(specimen_id)``

```
head(abdata)
```

	specimen_id	subject_id	actual_day_relative_to_boost				
1	1	1	-3				
2	1	1	-3				
3	1	1	-3				
4	1	1	-3				
5	1	1	-3				
6	1	1	-3				
	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex		
1	0	Blood	1	wP	Female		
2	0	Blood	1	wP	Female		
3	0	Blood	1	wP	Female		
4	0	Blood	1	wP	Female		
5	0	Blood	1	wP	Female		
6	0	Blood	1	wP	Female		
	ethnicity	race	year_of_birth	date_of_boost	dataset		
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset		
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset		
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset		
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset		
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset		

```

6 Not Hispanic or Latino White    1986-01-01    2016-09-12 2020_dataset
      age isotype is_antigen_specific antigen      MFI MFI_normalised
1 14311 days      IgE              FALSE    Total 1110.21154      2.493425
2 14311 days      IgE              FALSE    Total 2708.91616      2.493425
3 14311 days      IgG              TRUE      PT   68.56614      3.736992
4 14311 days      IgG              TRUE      PRN  332.12718      2.602350
5 14311 days      IgG              TRUE      FHA 1887.12263     34.050956
6 14311 days      IgE              TRUE      ACT   0.10000      1.000000
      unit lower_limit_of_detection
1 UG/ML                2.096133
2 IU/ML                29.170000
3 IU/ML                0.530000
4 IU/ML                6.205949
5 IU/ML                4.679535
6 IU/ML                2.816431

```

How many different antibody isotypes are there in this dataset?

```
length(abdata$isotype) #total number for isotypes
```

```
[1] 61956
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
#How many unique isotypes?
table(abdata$isotype)
```

```

IgE  IgG  IgG1  IgG2  IgG3  IgG4
6698 7265 11993 12000 12000 12000

```

```
#how many diff antigens?
table(abdata$antigen)
```

```

      ACT  BETV1      DT  FELD1      FHA  FIM2/3  LOLP1      LOS Measles      OVA
1970   1970   6318   1970   6712   6318   1970   1970   1970   6318
      PD1      PRN      PT      PTM  Total      TT
1970   6712   6712   1970   788   6318

```

Measles act as a control, not expected to see anything if someone gets pertussis vaccine. More likely to see higher levels of pertussis toxin if receiving pertussis vaccine.

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

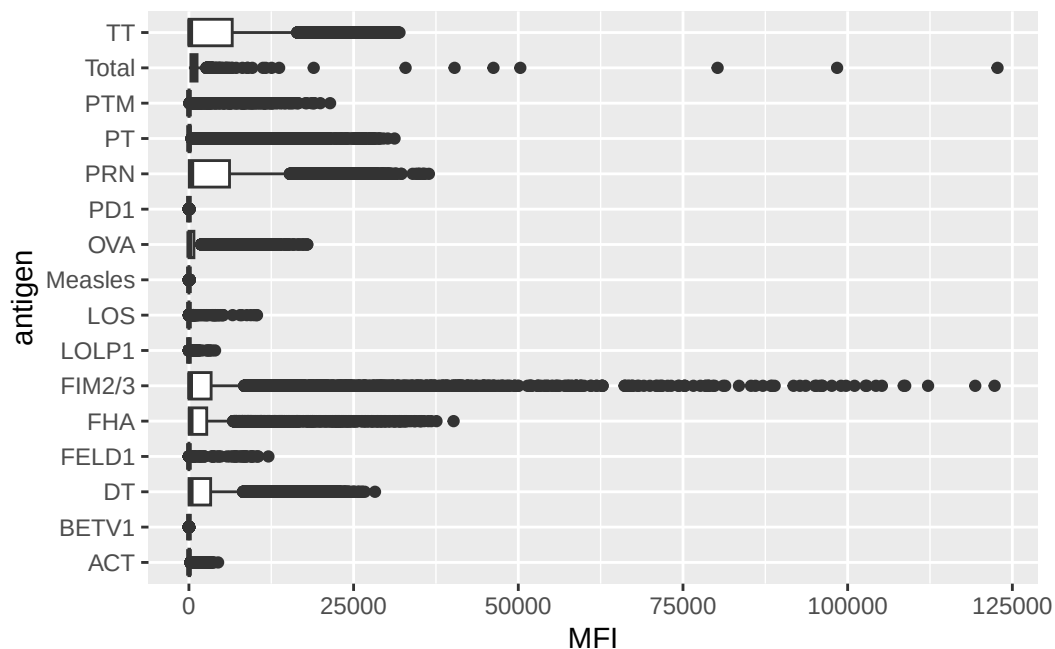
```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
      31520       8085       7301       15050
```

There are 4 different \$dataset values, and there are 15050 rows for the most recent 2023 dataset.

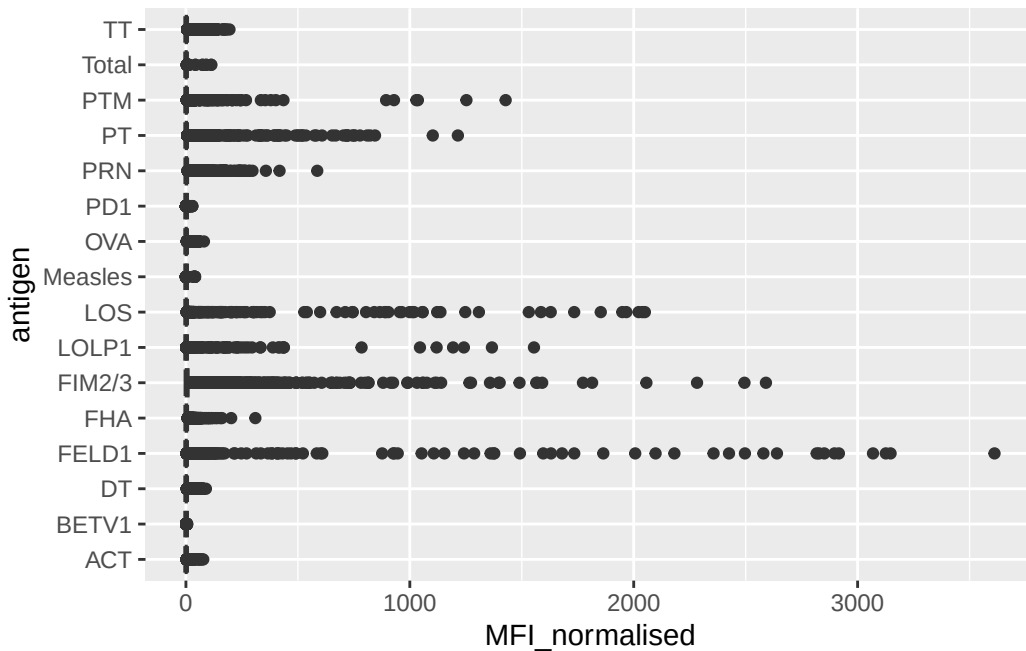
I want a plot of antigen levels across the whole dataset.

```
ggplot(abdata) +  
  aes (x=MFI, y= antigen) +  
  geom_boxplot()
```

Warning: Removed 1 row containing non-finite outside the scale range (`stat\_boxplot()`).



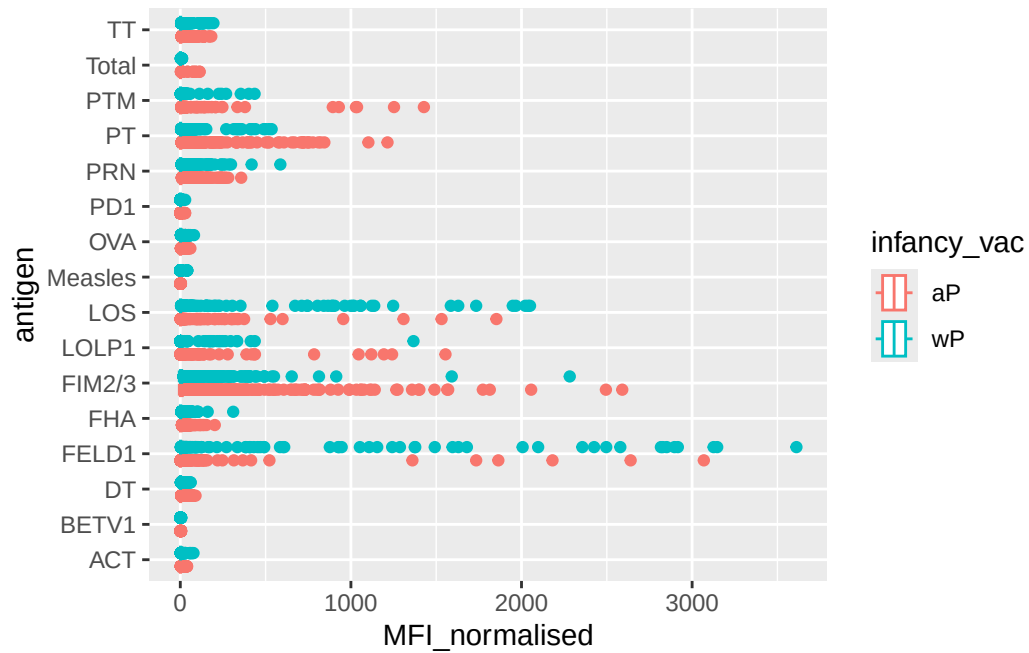
```
#Plot using MFI_normalised
ggplot(abdata) +
  aes (x=MFI_normalised, y= antigen) +
  geom_boxplot()
```



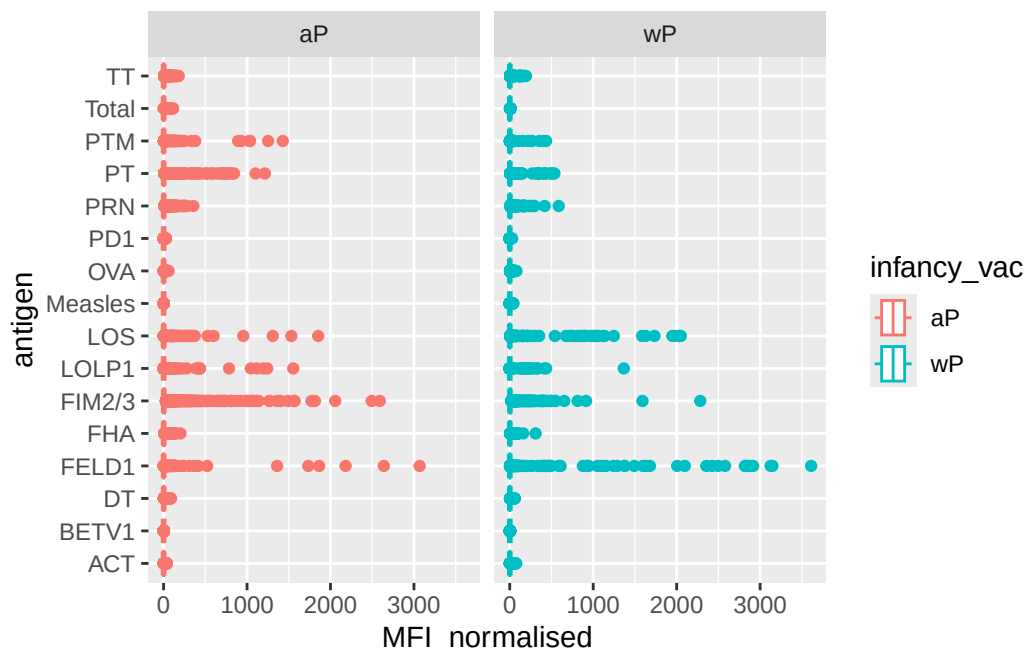
Antigens like FIM2/3, PT, FELD1 have a large range of values since they are the toxins in the pertussis vaccine. Others like Measles don't show much activity because they act as a control.

Q. Are the antibody levels different at the whole-dataset levels in wP vs aP individuals?

```
ggplot(abdata) +
  aes (x=MFI_normalised, y= antigen, col = infancy_vac) +
  geom_boxplot()
```



```
ggplot(abdata) +
  aes (x=MFI_normalised, y= antigen, col = infancy_vac) +
  geom_boxplot() +
  facet_wrap(~infancy_vac)
```



```
#facet_wrap pulls it into two separate columns
```

4. Examine IgG Ab titer levels

For this I need to select out just isotype G.

```
igg <- abdata |>
  filter(isotype == "IgG")
head(igg)
```

	specimen_id	subject_id	actual_day_relative_to_boost			
1	1	1	-3			
2	1	1	-3			
3	1	1	-3			
4	2	1	1			
5	2	1	1			
6	2	1	1			
	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex	
1	0	Blood	1	wP	Female	
2	0	Blood	1	wP	Female	
3	0	Blood	1	wP	Female	

4		1	Blood	2	wP	Female
5		1	Blood	2	wP	Female
6		1	Blood	2	wP	Female

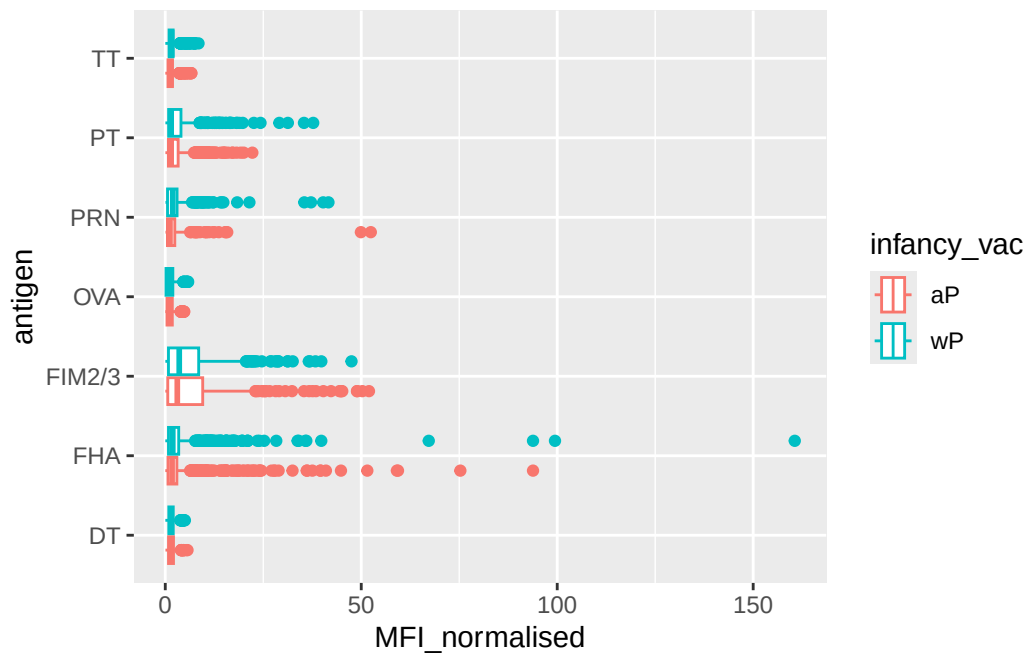
	ethnicity	race	year_of_birth	date_of_boost	dataset
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
4	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
5	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset
6	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset

	age	isotype	is_antigen_specific	antigen	MFI	MFI_normalised
1	14311 days	IgG	TRUE	PT	68.56614	3.736992
2	14311 days	IgG	TRUE	PRN	332.12718	2.602350
3	14311 days	IgG	TRUE	FHA	1887.12263	34.050956
4	14311 days	IgG	TRUE	PT	41.38442	2.255534
5	14311 days	IgG	TRUE	PRN	174.89761	1.370393
6	14311 days	IgG	TRUE	FHA	246.00957	4.438960

	unit	lower_limit_of_detection
1	IU/ML	0.530000
2	IU/ML	6.205949
3	IU/ML	4.679535
4	IU/ML	0.530000
5	IU/ML	6.205949
6	IU/ML	4.679535

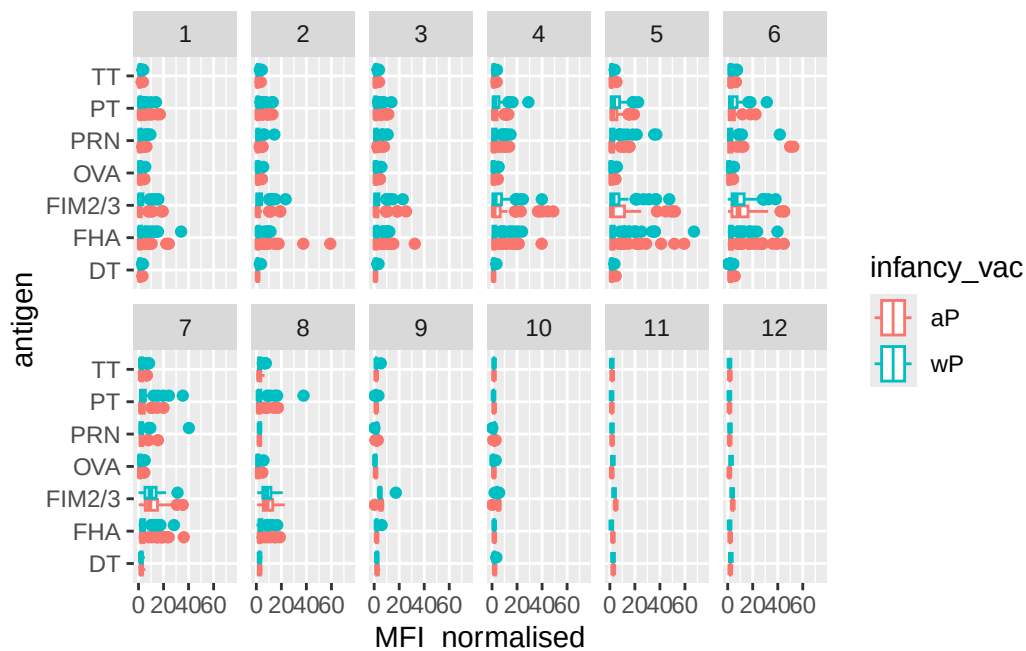
Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens

```
ggplot(igg) +
  aes (x=MFI_normalised, y= antigen, col = infancy_vac) +
  geom_boxplot()
```



```
ggplot(igg) +
  aes(MFI_normalised, antigen, col = infancy_vac) +
  geom_boxplot() +
  xlim(0,75) +
  facet_wrap(vars(visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).



Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

We see differences in level of IgG antibody titers recognizing antigens like FIM2/3, PT, FELD1 (have a large range of values) since they are the toxins present in the pertussis vaccines. Others like Measles don't show much activity because they are not in the vaccines and act as a control.

Digging in further to look at the time course of IgG isotype PT antigen levels across aP and wP individuals:

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")
abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
        y=MFI_normalised,
        col=infancy_vac,
        group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
```

```
labs(title="2021 dataset IgG PT",
      subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

